

Nion Rahaman Akash

B.Sc. in Mathematics | Postgraduate Diploma in Data Science

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Dhaka, Bangladesh

RESEARCH INTERESTS

Data Science | Machine Learning – Trustworthy and Reliable | Health Care | Natural Language Processing for Low-Resource Languages

EDUCATION

Postgraduate Diploma (PGD) in Data Science

United International University (UIU), Bangladesh • 2024 • CGPA 3.44/4.00

- Key areas: Python, Machine Learning, Deep Learning, SQL, Data Analysis, Data Visualization

B.Sc. in Mathematics

Dept. of Mathematics, University of Rajshahi, Bangladesh • 2016 – 2022 • CGPA 2.96/4.00

- Major theory courses: Theory of Statistics, Statistical Methods and Demography, Advanced Calculus, Vector and Tensor Analysis, Ordinary and Partial Differential Equations, Computer Programming (Fortran 90), Theory of Matrices, Operations Research, Real Analysis, Linear Algebra, Numerical Analysis, Number Theory, Discrete Mathematics and Programming with C

Applied Statistics using R

ISRT, University of Dhaka, Bangladesh • 36-hour intensive course

MAJOR PROJECTS

Diabetes Prediction using Machine Learning

- Built predictive models for diabetes risk using Logistic Regression, Random Forest, and a Keras/TensorFlow neural network
- Performed data preprocessing, feature engineering, model training, and evaluation
- Applied Python-based data science libraries for analysis and experimentation
- Achieved improved performance with ensemble and neural network approaches, reaching 75% accuracy

Other Machine Learning Projects

- **Car & Laptop Price Prediction** — regression pipelines (Linear, Ridge, Random Forest, Gradient Boosting) with full data-cleaning and cross-validated model comparison
- **Breast Cancer Classification** — Support Vector Machines (SVM) for tumor classification
- **Digit Recognition** — Random Forest and K-Nearest Neighbors (KNN) on handwritten digit data
- **Spam Detection & Human Sentiment Analysis** — NLP (Bag-of-Words, Naive Bayes) for text classification
- **Customer Segmentation** — K-Means Clustering and RFM Analysis for customer behavior grouping

Full code, write-ups, and results for these projects: github.com/Akash-Nion

CERTIFICATIONS

- R for Applied Statistics and Data Science — ISRT, University of Dhaka (36-hour intensive)
- Python for Data Science, AI & Development — Coursera

TECHNICAL SKILLS

- **Programming:** Python, R, C, Fortran, MATLAB
- **Machine Learning:** Regression, Classification, Clustering, Model Evaluation
- **Data Science Libraries:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn
- **Databases & Query Languages:** SQL, Neo4j, Cypher
- **Data Preparation:** Feature scaling, encoding techniques, outlier detection (IQR, Z-score)

EXTRACURRICULAR & LEADERSHIP ACTIVITIES

Volunteer, Transparency International Bangladesh (TIB)

Feb 2021 – Jan 2023

- Conducted primary data collection across diverse rural and urban households to support evidence-based social accountability research
- Promoted institutional transparency and ethical governance in public sectors (Health, Education) through targeted anti-corruption initiatives
- Facilitated information dissemination campaigns, translating complex governance protocols into accessible insights for non-technical citizens

Active Member, University of Rajshahi English Speaking Club

- Facilitated structured communication workshops, enhancing the public speaking and technical articulation skills of diverse peer groups
- Led collaborative learning sessions, fostering an environment of constructive feedback and continuous improvement in academic fluency

STANDARDIZED TEST

International English Language Testing System (IELTS)

28 Jan 2026

- Overall 6.5 (Listening 7.5, Reading 6.5, Writing 6.0, Speaking 6.0)

REFERENCES

Dr. MD. Sahadat Hossain

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